Expectation

1. Idea of expectation

A random variable (RV) takes a real value each time a random experiment is performed.

Expectation seeks to represent the average of these values over a large number of experiments.

2. Motivation (Discrete Cases)

Let:

- $X =$ discrete RV
- pmf p_X
- support E_X - This is the support of X. The set of all possible values that X can take.
- $\mathbb{E}[X]$ - Expectation (expected value) of the random variable X
  - $p_X(k)$ - This is the probability mass function (PMF).

Suppose we perform **N experiments.**

Let:

$N_k =$ # times $X = k$

Then empirically:

$$\frac{N_k}{N} \to p_X(k)$$

The sample average:

$$\frac{1}{N} \sum_{k \in E_X} k N_k = \sum_{k \in E_X} k \frac{N_k}{N}$$

Taking limit as $N \to \infty$:

$$\mathbb{E}[X] = \sum_{k \in E_X} k \, p_X(k)$$

K is every possible value k can take while performing N experiments

E_X -
Interpretation: Weighted average of values X takes

3. Definition

DISCRETE RV

If X is discrete with pmf p_X:

$$\mathbb{E}[X] = \sum_{k \in E_X} k \, p_X(k)$$

CONTINUOUS RV

If X is continuous with pdf $f_X(x)$:

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) \, dx$$

This is the **general definition of expectation.**

4. Examples

1. UNIFORM DISTRIBUTION

$U \sim$ Unif$[a, b]$

PDF:

$$f_U(x) = \frac{1}{b-a}, \quad a \le x \le b$$

Expectation:

$$\mathbb{E}[U] = \int_a^b x \frac{1}{b-a} dx = \frac{a+b}{2}$$

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Density
 |
 |    ___________
 |    |          |
 |___ |__________|________ x
     a           b
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Mean lies exactly at midpoint:

$(a+b)/2$

2. EXPONENTIAL DISTRIBUTION

$X \sim$ Exp(μ)

PDF:

$$f_X(x) = \mu e^{-\mu x}, \quad x \ge 0$$

$$\mathbb{E}[X] = \int_0^\infty x\mu e^{-\mu x} dx$$

Substitute:

$$t = \mu x$$

Then:

$$\mathbb{E}[X] = \frac{1}{\mu}$$

Note:

$$\int_0^\infty y e^{-y} dy = 1$$

(Gamma distribution with parameter 1)

3. BINOMIAL DISTRIBUTION

$$N \sim \text{Bin}(n, p)$$

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$$

$$\mathbb{E}[N] = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}$$

Using:

$$(1+x)^n = \sum_{k=0}^{n} \binom{n}{k} x^k$$

Differentiate:

$$n(1+x)^{n-1} = \sum_{k=0}^{n} k \binom{n}{k} x^{k-1}$$

After algebra:

$$\mathbb{E}[N] = np$$

4. NORMAL DISTRIBUTION

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$\mathbb{E}[X] = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} x e^{-(x-\mu)^2/(2\sigma^2)} dx$$

Substitute:

$y = x - \mu$

Then:

$$\mathbb{E}[X] = \mu + \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} y e^{-y^2/(2\sigma^2)} dy$$

Since integrand is odd function over symmetric limits:

$$\int_{-\infty}^{\infty} y e^{-y^2/(2\sigma^2)} dy = 0$$

Thus:

$\mathbb{E}[X] = \mu$

ADDITIONAL RESULTS

$N \sim \text{Geo}(p)$

$p_X(k) = (1-p)^{k-1} p, \quad k \geq 1$

$$\mathbb{E}[X] = \frac{1}{p}$$

$N \sim \text{Poi}(\lambda)$

$\mathbb{E}[N] = \lambda$

5. CAUCHY DISTRIBUTION

$X \sim$ Cauchy RV

$$f_X(x) = \frac{1}{\pi(1+x^2)}, \quad x \in \mathbb{R}$$

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)} dx$$

Splitting:

$$\int_0^{\infty} \frac{x}{\pi(1+x^2)} dx + \int_{-\infty}^{0} \frac{x}{\pi(1+x^2)} dx$$

Both integrals diverge:

$$= \infty - \infty$$

Expectation is undefined

6. PARETO DISTRIBUTION

$X \sim \text{Pareto}(x_0, \alpha)$

where:

$$x_0, \alpha > 0$$

Tail probability:

$$P(X > x) = \begin{cases} \left(\frac{x}{x_0}\right)^{-\alpha} & x \geq x_0 \\ 1 & x < x_0 \end{cases}$$

PDF:

$$f_X(x) = \begin{cases} \frac{\alpha}{x_0^{-\alpha}} x^{-(\alpha+1)} & x \geq x_0 \\ 0 & \text{otherwise} \end{cases}$$

Expectation:

$$\mathbb{E}[X] = \int_{x_0}^{\infty} \alpha x_0^{\alpha} x^{-(\alpha+1)} x \, dx$$

$$= \frac{\alpha}{\alpha - 1} x_0 \quad \text{if } \alpha > 1$$

$$= \infty \quad \text{if } \alpha \leq 1$$

Interpretation:

Infinite expectation for $\alpha \leq 1$ because tail probabilities decay polynomially, not exponentially.

Key Conceptual Takeaways

1. Expectation = long-run average
2. Discrete → summation
3. Continuous → integral
4. Symmetry can simplify integrals (Normal case)

5. Heavy-tailed distributions may have:
   - Infinite mean (Pareto)
   - Undefined mean (Cauchy)